

LegalEase: AI-Powered Legal Document Generator

Project Planning Phase

1. Team Details

Team Name: LegalEase: AI-Powered Legal Document generator

Team Members:

- 1. Bharathkumar A
- 2. Gurusankar S
- 3. Bikash Thaabaa R
- 4. Sabarivasan S

2. Introduction

The Project Planning phase defines how the LegalEase project will be organized, developed, tested, and completed. It identifies the major activities, responsibilities, resources, and expected outcomes of the project.

3. Project Objectives

- Develop an AI-powered legal document generation application.
- Provide a simple interface for entering legal document information.
- Integrate Google Gemini for document generation.
- Provide editable document previews.
- Support PDF and Word document downloads.
- Complete the project through systematic development and testing.

4. Project Activities

- Requirement Analysis – identify user and system requirements.
- Project Design – define architecture, modules, and data flow.
- Environment Setup – configure Python, virtual environment, and required packages.
- Frontend Development – create the Streamlit user interface.
- Backend Development – create the FastAPI API and request handling.
- AI Integration – connect the Gemini model for document generation.
- Document Processing – implement PDF and Word export.
- Testing – test generation, editing, API communication, and downloads.
- Documentation – prepare project reports and supporting documents.

5. Technology Planning

- Programming Language: Python
- Frontend: Streamlit
- Backend: FastAPI and Uvicorn
- AI Service: Google Gemini
- Document Processing: python-docx and PDF generation libraries
- Configuration: python-dotenv
- Development Tool: Visual Studio Code
- Version Control: GitHub

6. Team Responsibility Planning

- Team members will collaborate on requirement analysis, design, development, testing, and documentation.
- Frontend-related tasks include user interface and document preview.
- Backend-related tasks include API development and request validation.
- AI-related tasks include prompt preparation and Gemini integration.
- Testing and documentation will be carried out collaboratively.

7. Development Plan

- Phase 1 – Understand the problem and collect requirements.
- Phase 2 – Design the application architecture and workflow.
- Phase 3 – Set up the development environment.
- Phase 4 – Develop frontend and backend modules.
- Phase 5 – Integrate the Gemini AI module.
- Phase 6 – Implement document editing and export.
- Phase 7 – Test and fix errors.
- Phase 8 – Finalize documentation and project presentation.

8. Resource Planning

- Laptop or desktop computer
- Python development environment
- Internet connection
- Google Gemini API access
- Required Python packages
- GitHub repository for source-code management

9. Risk Planning

- AI API errors or service availability issues may affect document generation.
- Incorrect user input may produce incomplete document drafts.
- PDF formatting or special characters may require additional handling.
- API keys must be protected and should not be exposed publicly.
- Generated legal content should be reviewed before real-world use.

10. Testing Plan

- Test different document types such as NDA, Employment Contract, and Lease Agreement.
- Test valid and incomplete user inputs.
- Test FastAPI /generate endpoint.
- Test AI-generated document output.
- Test document editing.
- Test PDF and Word downloads.
- Check application errors and response handling.

11. Expected Outcome

The planned outcome is a working LegalEase application that accepts basic legal document information, generates a structured draft using Generative AI, allows the user to preview and edit the content, and provides downloadable document formats.

12. Project Planning Conclusion

The Project Planning phase provides a clear roadmap for completing LegalEase. By organizing activities, technologies, resources, responsibilities, testing, and risks, the team can proceed with development in a structured manner.